Econ 381: Introduction to Econometrics
Handout 8: Inference in Multiple Regression

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Where Handout 7 left us

Including the percentage of English learners, we estimated

$$\widehat{\text{TestScore}} = 686.03 - 1.10\,STR - 0.65\,PctEL.$$

The slope on STR compares class sizes **holding** $PctEL$ **fixed**.

The superintendent's next questions

How uncertain is that slope? Do school resources matter jointly? Which other district characteristics should we hold fixed?

Handout 6 gave us inference for one slope. We now use the same logic for multiple regression and extend it to **several coefficients at once**.

Source: Stock and Watson, `caschool.dta`; 420 California school districts. Robust standard errors.

Roadmap

We continue with the California school example:

1. **7.1: One coefficient**.
   Use familiar t-tests and confidence intervals.

2. **7.2: Joint hypotheses**.
   Use an F-test for several equalities at once.

3. **7.3: An equality involving two coefficients**.
   Test the equality directly in Stata.

4. **7.4: Joint confidence sets**.
   Represent uncertainty about a pair of coefficients.

5. **7.5–7.6: Controls and model choice**.
   Choose a benchmark model and assess plausible alternatives.

Section 7.1

Hypothesis tests and confidence intervals for one coefficient

One coefficient: confidence intervals and tests

In multiple regression, the slope β_j describes a change in X_j **holding the other included regressors fixed**.

The same confidence interval as in Handout 6

A large-sample 95% confidence interval is

$$\widehat{\beta}_j \pm 1.96 \text{ SE}(\widehat{\beta}_j).$$

Use the heteroskedasticity-robust standard error.

To test $H_0 : \beta_j = b_{j,0}$ against $H_1 : \beta_j \neq b_{j,0}$ at 5%,

- reject if the 95% interval excludes $b_{j,0}$,
- equivalently, reject if $|t| > 1.96$, where $t = (\widehat{\beta}_j - b_{j,0}) / \text{SE}(\widehat{\beta}_j)$,
- or reject if the p-value is below 0.05.

Under the least squares assumptions, these rules control the false rejection rate at approximately 5% in large samples.

Stata: inference for the class size slope

```
use "caschool.dta", clear
reg testscr str el_pct, robust
```

Stata output (coefficient table):

```
------------------------------------------------------------------------------
             |               Robust
     testscr | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |  -1.101296   .4328472    -2.54   0.011     -1.95213   -.2504616
      el_pct |  -.6497768   .0310318   -20.94   0.000     -.710775   -.5887786
       _cons |   686.0322   8.728224    78.60   0.000     668.8754     703.189
------------------------------------------------------------------------------
```

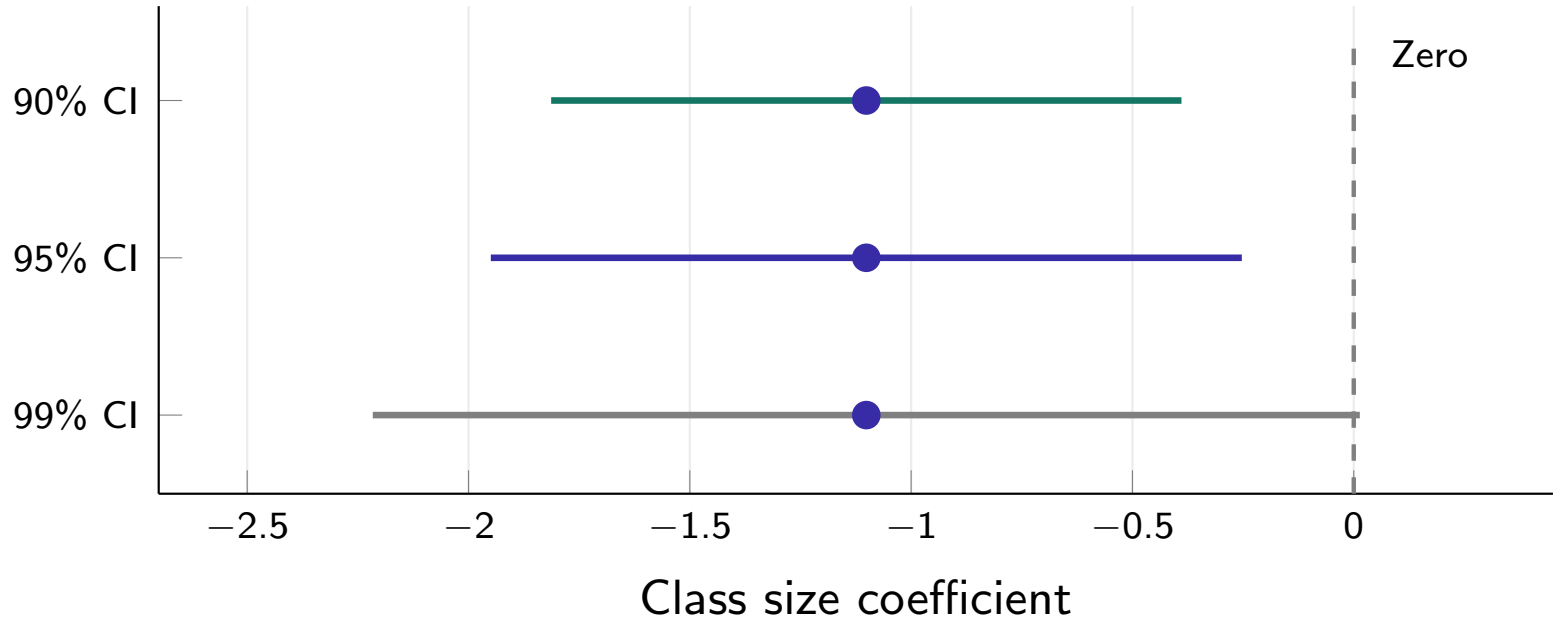
- `el_pct` is the dataset's name for *PctEL*.
- The t and p columns test **coefficient** $= 0$ by default.
- `robust` changes standard errors, not the OLS estimates.

Stata uses finite-sample t reference values here. Its intervals can differ slightly from the large-sample 1.96 formula.

Source: Stock and Watson, `caschool.dta`; 420 California school districts. Robust standard errors.

Reading the class size result

Holding *PctEL* fixed, one more student per teacher is associated with **1.10 fewer test score points** (SE = 0.433).



The 95% interval excludes zero. The 99% interval includes it.

$$|t| = 2.54, \qquad p = 0.011: \quad \textbf{reject at 5\%, but not at 1\%.}$$

Plotted intervals use large-sample cutoffs. Stata's finite-sample intervals differ slightly. A causal interpretation still needs credible controls.

Section 7.2

Tests of joint hypotheses

Class size and school spending

Hiring more teachers can require cutting spending on other school inputs. Add spending per pupil, *Expn* (in dollars), to the model:

$$\text{TestScore}_i = \beta_0 + \beta_1 STR_i + \beta_2 Expn_i + \beta_3 PctEL_i + u_i.$$

- **Class size with spending fixed**: does *STR* matter after holding total spending and *PctEL* fixed?

$$H_0 : \beta_1 = 0.$$

- **School resources jointly**: do class size and spending affect test scores, holding *PctEL* fixed?

$$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0.$$

The second question requires **one test of both equalities together**.

Stata: add spending per pupil

```
reg testscr str expn_stu el_pct, robust
```

Stata output (coefficient table):

```
------------------------------------------------------------------------------
             |               Robust
     testscr | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |  -.2863992   .4820728    -0.59   0.553    -1.234002     .661203
    expn_stu |   .0038679   .0015807     2.45   0.015     .0007607    .0069751
      el_pct |  -.6560227   .0317844   -20.64   0.000    -.7185008   -.5935446
       _cons |   649.5779   15.45834    42.02   0.000     619.1917    679.9641
------------------------------------------------------------------------------
```

- *STR*: do not reject a zero slope at 5% ($p = 0.553$).
- Spending: reject a zero slope at 5% ($p = 0.015$). An additional \$1,000 is associated with about 3.87 more points, holding the other regressors fixed.

Failure to reject $\beta_1 = 0$ does **not imply** that $\beta_1 = 0$. If the true slope is nonzero, this is a Type-II error.

Source: Stock and Watson, `caschool.dta`; 420 California school districts. Robust standard errors.

The joint null and its alternative

For the school resources question,

$$H_0 : \beta_1 = 0 \textbf{ and } \beta_2 = 0,$$
$$H_1 : \beta_1 \neq 0 \textbf{ or } \beta_2 \neq 0 \quad \text{(or both)}.$$

The null imposes **two restrictions**, or two separate equalities.

A naive procedure

Run two individual 5% t-tests. Reject the joint null if either one rejects.

Even if both population slopes are zero, we have **two possibilities** for a false rejection. The overall chance of a mistake can exceed 5%.

We need a test calibrated for the **joint question**.

Two chances for a false rejection

Suppose the two individual 5% tests are **independent** and both population slopes are zero.

Individual test outcomes	Probability	Percentage
Neither rejects	0.95^2	90.25%
Exactly one rejects	$2(0.95)(0.05)$	9.50%
Both reject	0.05^2	0.25%
At least one rejects	$0.095 + 0.0025$	9.75%

The Type-I error of the naive procedure

Rejecting whenever **at least one** individual test rejects gives

$$9.50\% + 0.25\% = 9.75\% > 5\%.$$

In general the two tests (or slope estimates) are correlated, so the error rate also depends on that correlation. A joint test accounts for this.

The F-statistic: evidence against the joint null

The joint null says that the population coefficient pair is $(0, 0)$. The sample estimates will usually differ from zero, even when that null is true.

The idea of the F-test

Are the estimated slope coefficients, if combined together, unusually far from the hypothesized values relative to their sampling uncertainty?

- F is nonnegative. A **larger** F gives stronger evidence against the joint null.
- The calculation accounts for the precision of each estimate and how the estimates vary together across samples.
- After a regression with `robust`, Stata can compute a **robust joint test** with the command `test`.

One test assesses the entire set of equalities at the chosen significance level.

Why larger F means stronger evidence (optional)

For two zero restrictions, when the coefficient estimates are independent,

$$\boxed{F = \frac{t_1^2 + t_2^2}{2}.}$$

It is the **average of two squared t-statistics**.

t_1	t_2	F	Interpretation
1	1	1.0	Both estimates are close to zero relative to their SEs.
2	1	2.5	One estimate is farther from zero.
2	2	4.0	Both estimates are farther from zero.
2	-2	4.0	Either sign can provide evidence against zero.

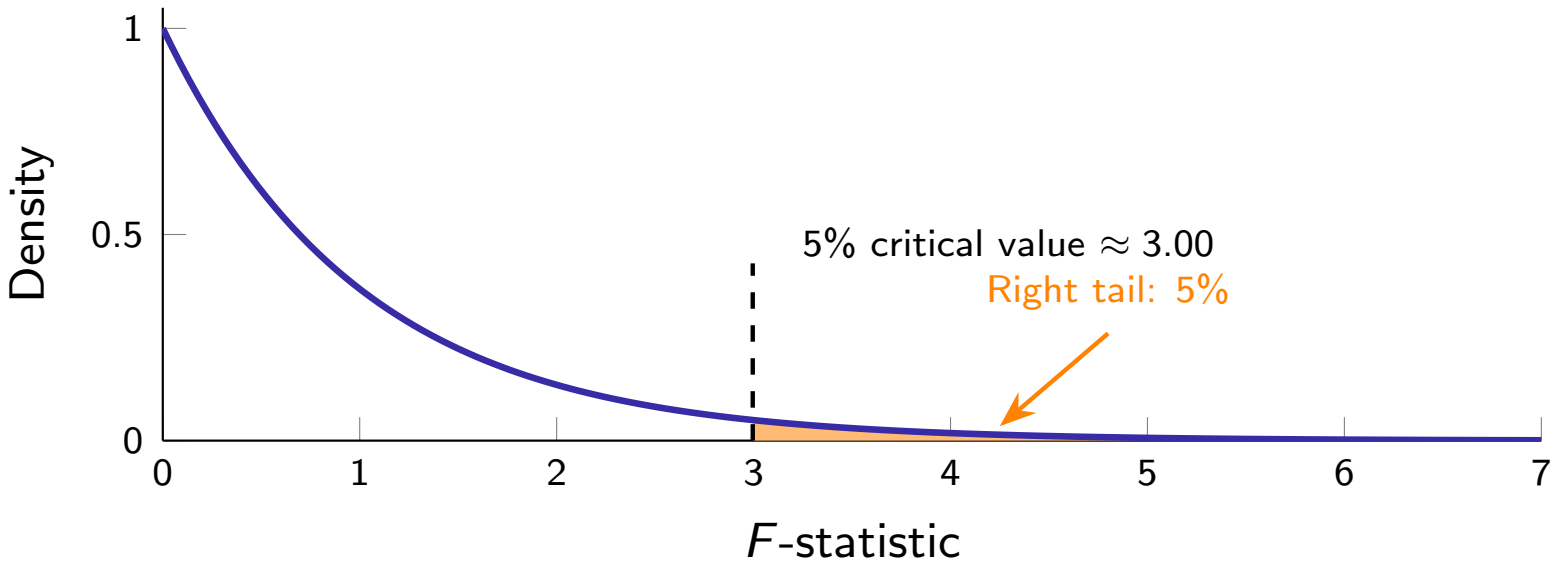
- Squares cannot be negative, so $F \geq 0$.
- Increasing either $|t_j|$, holding the other fixed, increases F.

With correlated estimates, Stata adjusts the calculation. The formula above applies only to the independent case.

The F reference distribution

Under a joint null with **two restrictions**, F-statistic has a large-sample limiting distribution $F_{\mathbf{2},\infty}$.

Like $N(0,1)$, $F_{q,\infty}$ is a **limiting distribution** of the F-statistic in a large sample. It is designed for joint tests, with q equality restrictions in the null.



Large F values provide evidence against the null. The p-value uses the **right tail**, even when the alternative allows positive or negative slopes.

Two rules for an F-test

For $H_0 : \beta_1 = \beta_2 = 0$, choose a significance level from $\{10\%, 5\%, 1\%\}$ first.

1. F-**statistic rule**: reject if F exceeds the critical value.

Significance level	10%	5%	1%
$F_{2,\infty}$ critical value	2.30	3.00	4.61

2. p-**value rule**: reject if the reported p-value is below that significance level.

A small p-value means that it is unusual to obtain an F-statistic at least as larger as the observed one **if the joint null were true**.

In practice, use the software's reported p-value for F-tests.

Stata: test school resources jointly

```
reg testscr str expn_stu el_pct, robust
test str expn_stu
```

Stata output:

```
 ( 1)  str = 0
 ( 2)  expn_stu = 0

       F(  2,   416) =    5.43
            Prob > F =    0.0047
```

- $5.43 > 4.61$, and $0.0047 < 0.01$: reject the joint null even at 1%.
- There is evidence that **at least one** of the two slopes is nonzero, holding *PctEL* fixed. This does not establish that both are nonzero.

Sidenote: Stata uses a slightly different reference distribution $F_{2,416}$, based on two equalities and $420 - 4 = 416$ residual degrees of freedom. Our critical value table uses its large-sample approximation $F_{2,\infty} \approx F_{2,416}$

Source: Stock and Watson, `caschool.dta`; 420 California school districts. Robust standard errors.

Counting restrictions

Let q be the number of **nonredundant equalities** in the null. The large-sample reference is $F_{q,\infty}$.

Null hypothesis	q
$\beta_1 = 0$	1
$\beta_1 = 0$ and $\beta_2 = 0$	2
$\beta_1 = \cdots = \beta_k = 0$ (all k slopes)	k

The test in the regression header (see next slide)

It tests **all slopes** of all regressors, jointly. The intercept remains unrestricted.

For the model with *STR*, spending, and *PctEL*, there are **three** slope restrictions. For just the two resource slopes, use `test str expn_stu` after the regression.

Stata: the regression header tests all slopes

```
reg testscr str expn_stu el_pct, robust
```

```
Linear regression                               Number of obs     =        420
                                                F(3, 416)         =     147.20
                                                Prob > F          =     0.0000
                                                R-squared         =     0.4366
                                                Root MSE          =     14.353

------------------------------------------------------------------------------
             |               Robust
     testscr | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |  -.2863992   .4820728    -0.59   0.553    -1.234002     .661203
    expn_stu |   .0038679   .0015807     2.45   0.015     .0007607    .0069751
      el_pct |  -.6560227   .0317844   -20.64   0.000    -.7185008   -.5935446
       _cons |   649.5779   15.45834    42.02   0.000     619.1917    679.9641
------------------------------------------------------------------------------
```

The highlighted header tests $H_0 : \beta_{STR} = \beta_{Expn} = \beta_{PctEL} = 0$. With $p < 0.01$, we reject that all three slopes are zero.

A displayed p-value of `0.0000` means very small, not exactly zero.

Source: Stock and Watson, `caschool.dta`; 420 California school districts. Robust standard errors.

Section 7.3

One equality involving multiple coefficients

The number of equalities determines the test

The subscript q of $F_{q,\infty}$ counts the number of **equality restrictions** in the null, rather than the coefficients involved in.

Null hypothesis	Number of Restrictions
$\beta_1 = \beta_2$	1
$\beta_1 = 0$ **and** $\beta_2 = 0$	2

A hypothesis on comparison in test score points

Does reducing STR by one student per teacher have the same predicted score gain as increasing spending by \$1,000 per pupil?

$$\underbrace{-\beta_{STR}}_{\text{one-student reduction}} = \underbrace{1000\beta_{Expn}}_{\text{\$1,000 more spending}} .$$

This is a null hypothesis written as **one equality**. Both gains from STR and Expn hold the other regressors fixed.

Stata: test the equality directly

The estimated score gains are 0.286 pts from reducing *STR* by one and 3.868 pts from \$1,000 more spending per pupil. These are **sample estimates**. The test asks whether the corresponding **population gains** are equal.

```
reg testscr str expn_stu el_pct, robust
test -str = 1000*expn_stu
```

Stata output:

```
 ( 1)  - str - 1000*expn_stu = 0

       F(  1,   416) =    3.63
            Prob > F =    0.0574
```

- $p = 0.0574 > 0.05$: do not reject equal score gains at 5%.
- At 10%, we reject. The significance level affects the decision.

Failure to reject at 5% does not establish equality of **population** gains.

Source: Stock and Watson, `caschool.dta`; 420 California school districts. Robust standard errors.

Stata: estimate the difference and its interval

`lincom` estimates the difference in predicted score gains:

gain from smaller classes $-$ gain from more spending.

```
lincom -str - 1000*expn_stu
```

Stata output (coefficient table):

```
------------------------------------------------------------------------------
     testscr | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |  -3.581503     1.8793    -1.91   0.057    -7.275611    .1126062
------------------------------------------------------------------------------
```

- The estimated difference is about -3.58 test score points.
- Its 95% interval includes zero, agreeing with the 5% F-test.

For one restriction, $F = t^2$. So the F-test from `test` command agrees with the t-test from `lincom` command.

Section 7.4

Confidence sets for multiple coefficients

Uncertainty about a pair of coefficients

A confidence interval gives plausible values for **one coefficient**. A joint confidence set gives plausible values for a **vector** (b_1, b_2).

Reversing the logic from Handout 6

For one coefficient, we used a confidence interval to construct a test. Here, we construct a confidence set by collecting all pairs a 5% F-test does NOT reject.

1. Choose a candidate pair (b_1, b_2).
2. Test the two equalities $\beta_1 = b_1$ and $\beta_2 = b_2$ together.
3. Keep (b_1, b_2) if the test does not reject it.
4. Repeat Steps 1-3 across all $(b_1, b_2) \in \mathbb{R}^2$.

Across repeated samples, the constructed set contains the **true pair** (β_1, β_2) approximately 95% of the time.

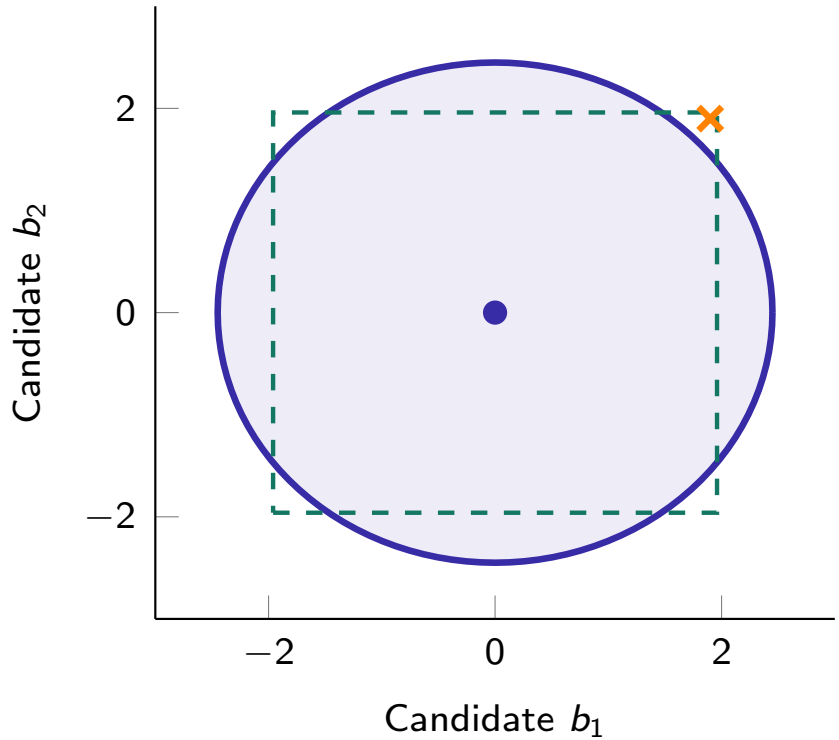
Separate intervals and a joint confidence set

Two intervals: invalid as a 95% joint set

- Each individual interval covers one true coefficient about 95% of the time.
- They need not cover **both** true coefficients 95% of the time.

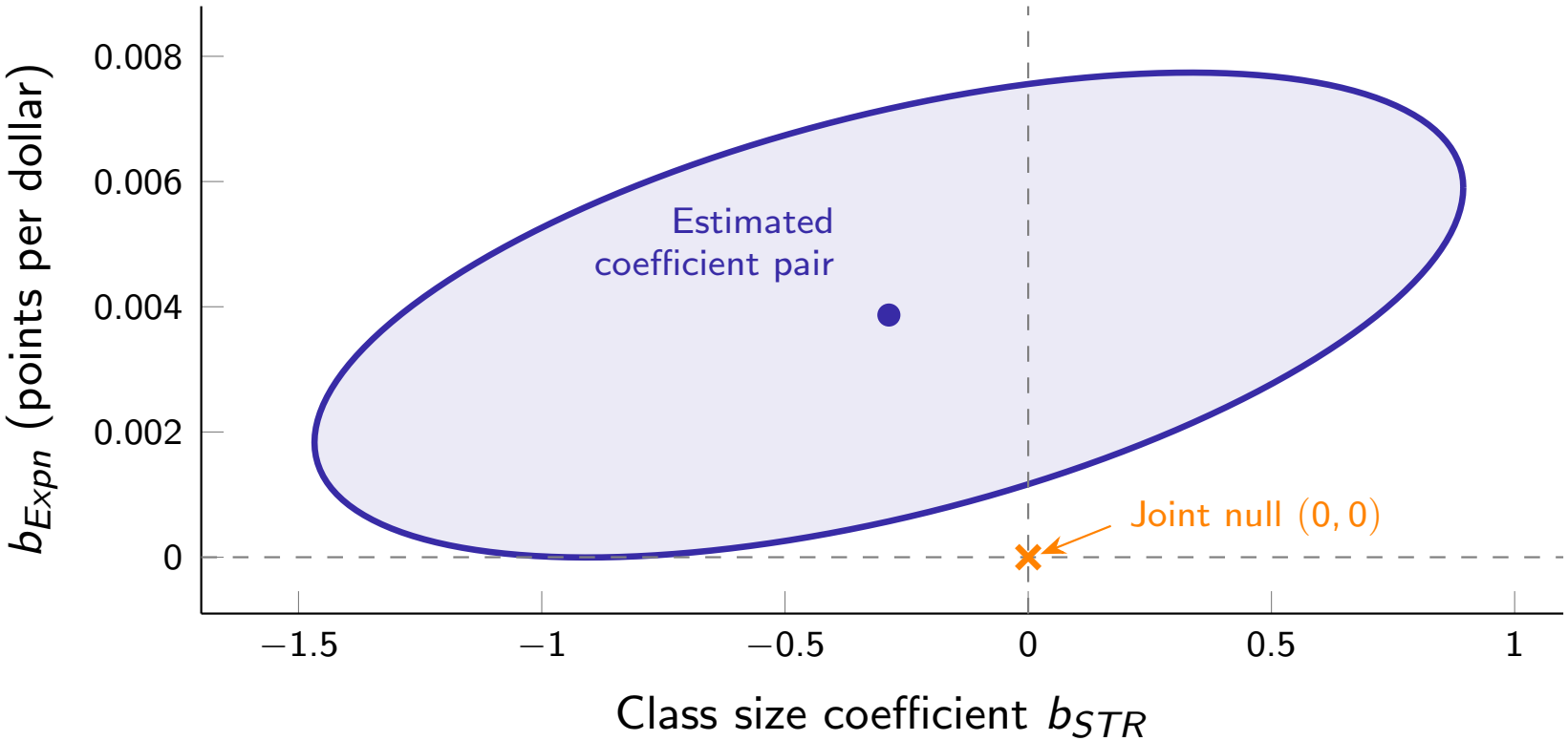
A valid 95% joint confidence set

Calibrates coverage for the pair.
With two coefficients, the usual F-test gives an ellipse.



Schematic in standardized units. Dashed square: pairs inside both individual intervals. Blue region: joint set. Orange pair: passes each individual test but fails the joint test. The two regions differ.

A joint confidence set for school resources



The center is located at the estimated coefficients. The width and tilt reflect uncertainty and how the estimates vary together.

The joint null $(0, 0)$ lies **outside** the 95% set: we reject that both resource slopes are zero at 5%.

Source: Stock and Watson, `caschool.dta`; 420 California school districts. Robust standard errors.

Section 7.5–7.6

Control variables and regression specification

The variable of interest and the controls

The research question determines the role of each variable. For the causal effect of class size:

Role	Variable	Purpose
Variable of interest	*STR*	Estimate the effect of changing class size.
Controls	*PctEL*, subsidized lunch share	Account for language and family background differences across districts.

A control can matter directly for test scores, or serve as a proxy for an omitted factor.

The aim is to make class size **as good as randomly assigned** among districts with the same observed characteristics.

Under **conditional mean independence**, the control's own coefficient need not be causal, even when the *STR* slope is.

What makes controls effective?

A useful control measures an omitted determinant of scores, or serves as a good proxy for one.

Control	The district difference it helps account for
PctEL (`el_pct`)	Non-native speakers difference. Compare districts with the same English learner share.
Lunch eligibility (`meal_pct`)	Family disadvantage. Compare districts with similar low-income shares.

The assumption we want after holding these controls fixed

STR is unrelated to the **remaining** omitted determinants of scores, on average:

$$E(u_i \mid STR_i, W_i) = E(u_i \mid W_i).$$

Here W contains the both controls variables. They do not guarantee this assumption. It needs outside knowledge and an appropriate regression specification.

Stata: a benchmark with language and family background

```
reg testscr str el_pct meal_pct, robust
```

Stata output (coefficient table):

```
------------------------------------------------------------------------------
             |               Robust
     testscr | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |  -.9983092   .2700799    -3.70   0.000    -1.529201   -.4674178
      el_pct |  -.1215733   .0328317    -3.70   0.000      -.18611   -.0570366
    meal_pct |  -.5473456   .0241072   -22.70   0.000    -.5947328   -.4999583
       _cons |     700.15    5.56845   125.74   0.000     689.2042    711.0958
------------------------------------------------------------------------------
```

`meal_pct` is the percentage eligible for subsidized lunches, a measure related to low family income.

- The class size slope is about -1.00.
- The lunch-share coefficient does NOT reflect the causal effect of lunch-program policy (Why?)

Source: Stock and Watson, `caschool.dta`; 420 California school districts. Robust standard errors.

Choosing the benchmark model

1. **Specify the causal question**.
   What effect does a change in *STR* have on scores?

2. **Identify possible confounders**.
   Which omitted determinants of scores also vary with class size? Family resources and language background are possible candidates.

3. **Choose controls with a substantive rationale**.
   Use available measures that make comparisons more credible. Here, *PctEL* and subsidized lunch share define a benchmark.

Ask whether remaining district differences on test scores could still be related to STR. Do not select controls only because they are statistically significant.

The benchmark is a reasonable starting point. It is not a guarantee that all omitted variable bias has disappeared.

Stata: plausible alternative controls

Use another measure of family disadvantage, `calw_pct`: the percentage qualifying for public income assistance.

```
* Benchmark: language and subsidized lunch share
reg testscr str el_pct meal_pct, robust
estimates store base

* Alternative measure of family disadvantage
reg testscr str el_pct calw_pct, robust
estimates store welfare

* Both measures
reg testscr str el_pct meal_pct calw_pct, robust
estimates store both

estimates table base welfare both, b(%9.3f) ///
    se(%9.3f) stats(N r2)
```

Compare the *STR* estimates and their uncertainty across these models. Keep the same sample when comparing specifications.

Stata output: comparing the specifications

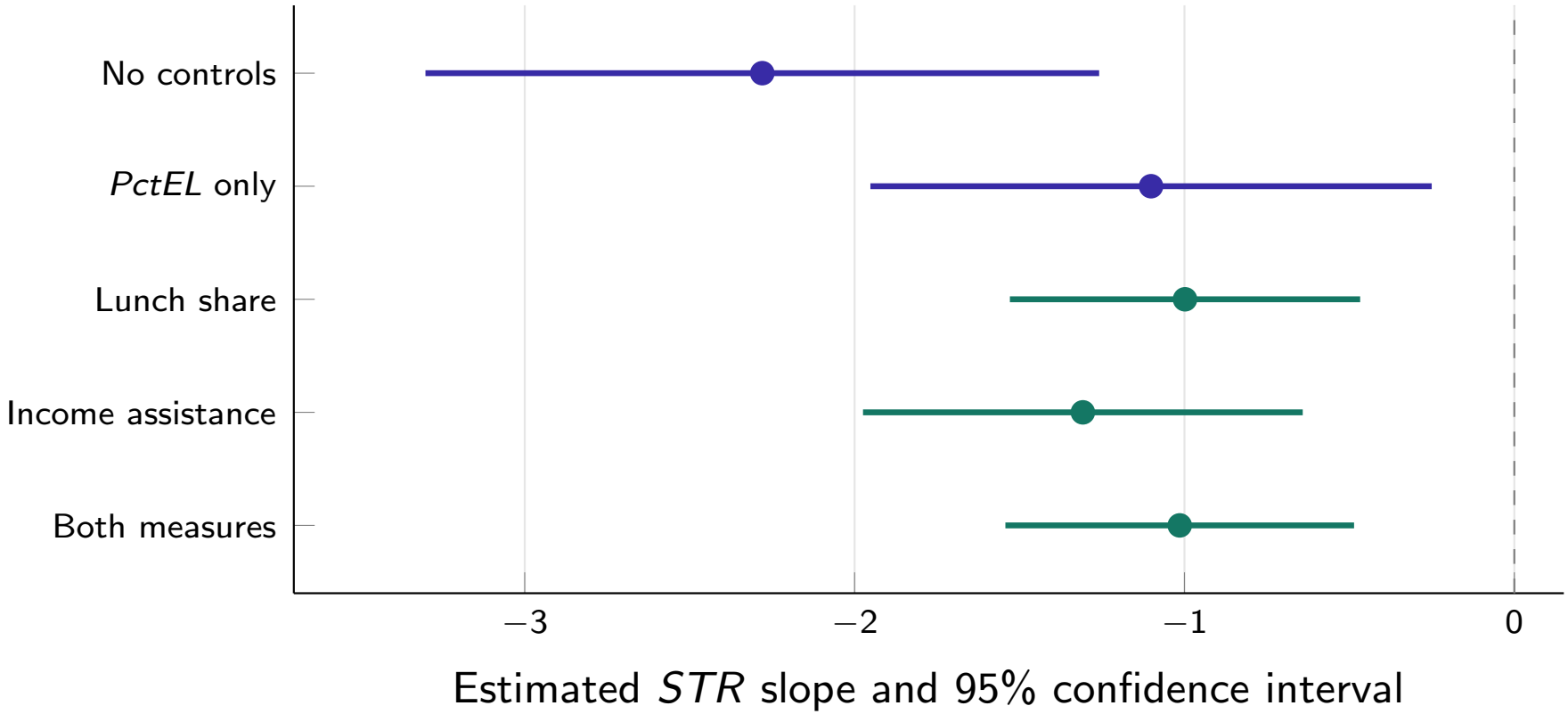
Variable	base	welfare	both
str	-0.998	-1.308	-1.014
	0.270	0.339	0.269
el_pct	-0.122	-0.488	-0.130
	0.033	0.030	0.036
meal_pct	-0.547		-0.529
	0.024		0.038
calw_pct		-0.790	-0.048
		0.068	0.059
_cons	700.150	697.999	700.392
	5.568	6.920	5.537
N	420	420	420
r2	0.775	0.629	0.775

Legend: b/se

Robust standard errors appear directly below coefficients. The *STR* slope remains negative, around -1.0 to -1.3.

Stable estimates of STR slope support the benchmark, but do not prove conditional independence.

How the class size estimate changes with controls



The bottom three models all include *PctEL*. Controlling for observed background differences moves the simple slope substantially. Remaining bias is still possible.

Source: Stock and Watson, `caschool.dta`; 420 California school districts. Robust standard errors.

Reporting regression results

A regression table lets readers compare specifications quickly.

- Label the outcome, included regressors, and units clearly.
- Report coefficients with robust standard errors, usually in parentheses.
- Report the sample size and measures of fit.
- State the null for any joint F-test and report its p-value.
- If using significance stars, define the levels in a table note.

Explain what the coefficient of interest means **in the units of the data** and which other variables the comparison holds fixed.

Report the benchmark and credible alternative designs so readers can assess both sampling uncertainty and specification sensitivity.

A regression table in the textbook format

Dependent variable: average district test score

Regressor	(1)	(2)	(3)	(4)	(5)
Student–teacher ratio	-2.280^{**}	-1.101^{*}	-0.998^{**}	-1.308^{**}	-1.014^{**}
	(0.519)	(0.433)	(0.270)	(0.339)	(0.269)
English learners (%)		-0.650^{**}	-0.122^{**}	-0.488^{**}	-0.130^{**}
		(0.031)	(0.033)	(0.030)	(0.036)
Lunch eligibility (%)			-0.547^{**}		-0.529^{**}
			(0.024)		(0.038)
Income assistance (%)				-0.790^{**}	-0.048
				(0.068)	(0.059)
Intercept	698.933^{**}	686.032^{**}	700.150^{**}	697.999^{**}	700.392^{**}
	(10.364)	(8.728)	(5.568)	(6.920)	(5.537)
SER	18.581	14.464	9.080	11.654	9.084
R^2	0.051	0.426	0.775	0.629	0.775
n	420	420	420	420	420

Robust SEs in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$ (two-sided tests). A blank means the regressor is omitted. Column (3) is the benchmark.

Source: Stock and Watson, Table 7.1. Estimates and robust standard errors recomputed using `caschool.dta`.

Fit, significance, and causality answer different questions

Evidence	What it addresses
High R^2	How much sample variation in scores the regressors explain.
Small p-value	Evidence against a stated coefficient restriction, given the model and assumptions.
Credible controls	Whether the class size comparison accounts adequately for omitted determinants of scores.

For causal analysis, a higher R^2 alone is not a reason to prefer a model.

More flexible specifications can help represent relationships with controls.
Next, Chapter 8 allows nonlinear models using the regression tools.

New notions and methods in Handout 8

1. **One coefficient**: use the familiar robust SE, t-test, and confidence interval, holding other regressors fixed.
2. **Joint hypotheses**: use one F-test for the stated equalities. Separate 5% t-tests need not control the overall error rate at 5%.
3. **Count restrictions**: $\beta_1 = \beta_2$ is one equality. Use `test` or `lincom` for an equality or a difference.
4. **Joint confidence sets**: retain all nonrejected coefficient pairs. The usual F-test gives an ellipse for two coefficients.
5. **Model choice**: justify the controls, choose a benchmark, and examine plausible alternatives. Fit and significance do not establish causality.